

SOFTWARE REQUIREMENTS SPECIFICATION

Milestone 4 (Final Milestone)

MarketMind AI – Small Business Sales Intelligence Platform

Team Size: 5 Interns | Duration: 7 Days | Mode: Parallel, Domain-wise Development

Written in Simple, Layman-Friendly Language — Uses Only Free / Non-Paid Tools

Focus: Final Testing, Zero-Cost Deployment, Full Integration & Documentation

Prepared for: Intern Training Cohort

Document Version: 1.0

1. Introduction

1.1 Purpose

This document explains, in simple everyday language, exactly what each of the five interns should do — day by day, for 7 days — to complete Milestone 4, the FINAL milestone of the MarketMind AI – Small Business Sales Intelligence Platform. Every day has a plain-English explanation of the task plus a checklist of checkpoints so interns can tick things off as they finish, without needing to be an expert to understand what is being asked.

1.2 Recap: What Milestones 1, 2 & 3 Already Built

Milestone 1 built the foundation: login with roles, sales upload, inventory tracking, a basic dashboard, a simple sales forecast, and a Docker-based local setup. Milestone 2 added full Invoice Management and four AI-powered insights (Customer Segmentation, Churn Risk, Product Recommendation, and Anomaly Detection) with matching dashboard screens. Milestone 3 added Notifications, an upgraded forecasting model, measurable accuracy improvements to every AI feature, automated testing, and full local integration — while intentionally holding back live deployment until now. The platform today is a complete, feature-rich shop that has only ever been tested on the team's own laptops.

1.3 Milestone 4 Objective (In Simple Words)

Milestone 4 is about proving the shop works in the real world, not adding new rooms to it. This directly matches the original project plan's Week 7 & 8 goal: perform final testing, deploy the application using Docker and free cloud platforms, prepare documentation, and demonstrate the complete platform. By Day 7, MarketMind AI should be reachable by anyone through a single free, public web link, with every feature from Milestones 1–3 working correctly on that live link, backed by final documentation and a rehearsed demonstration — at zero cost to the team.

1.4 Golden Rules for This Milestone

- **Free Tools Only:** every tool and hosting service used must be free — the final deployed platform must cost the team nothing to run.
- **Same Dataset, Same Scope:** this milestone reuses the exact same dataset seeded in Milestone 1 and grown naturally through Milestones 2 & 3. No new sample data, and no new features beyond what the original project plan and previous milestones already defined — Milestone 4 is about testing, deployment, stability and documentation, not new functionality.
- **Deployment Happens Now:** unlike Milestone 3, where deployment was intentionally only drafted, Milestone 4 is where the team actually executes that deployment to free hosting.
- **Stability Over Sophistication:** if a choice must be made between adding something clever and making the existing platform reliable and demo-ready, always choose reliability.

2. Free Tools Checklist (No Paid Tool Required Anywhere)

Before Day 1 begins, make sure every intern has access to the free tools below relevant to their domain. Most of these were already used in earlier milestones and are simply being put to full use now.

What It's Needed For	Free Tool Option(s)
Writing code	VS Code (completely free)
Saving & sharing code	Git + GitHub (free for public/private small repos)
Testing APIs manually	Postman (free plan) or the Thunder Client extension in VS Code (free)
Automated tests (already built in Milestone 3)	pytest / Jest / React Testing Library — reused, not rebuilt
Database	The same free-tier cloud database already set up in Milestone 2/3 (Supabase/Neon/ElephantSQL) — reused, not recreated
Deploying the Backend & AI services	Render, Railway or Fly.io — free tier
Deploying the Frontend	Netlify, Vercel or GitHub Pages — free tier
Checking the live site is up	UptimeRobot (free tier)
Simple load testing	Apache Bench (ab) — free, comes with most systems
Running everything together locally (for fallback/demo backup)	Docker Desktop (free) with docker-compose
CI/CD	GitHub Actions (free tier), already configured in Milestone 3
Final documentation	Google Docs / Markdown in the GitHub repo — both free

3. Team Structure & Domain Allocation (Same 5 Domains as Milestones 1–3)

Each intern continues to own the same domain from the previous milestones and now brings it to a stable, deployed, final state. Working in the same domain avoids relearning and keeps parallel work smooth right through to the finish line.

Intern	Role / Domain	Architecture Layer Owned	Milestone 4 Focus
Intern 1	Backend & Database Developer	Data Ingestion + Storage Layer	Final testing, error hardening, and live deployment of all backend APIs, without altering the underlying dataset.
Intern 2	Security & API Gateway Developer	Application Gateway & Security Layer	Security configuration verified for the live/public environment (CORS, secrets, RBAC), confirmed by automated and manual testing on the live link.
Intern 3	Frontend & Dashboard Developer	Presentation Layer	Deployed, responsive, error-resilient frontend connected to the live backend and AI services, tested across every screen.
Intern 4	AI/ML Engineer	AI Analytics Engine	All five AI features tested, packaged, deployed live, and documented with final performance metrics — using only the existing dataset.
Intern 5	Database & DevOps Engineer	Storage Layer + Infrastructure Layer	Complete free-tier deployment of backend, AI services, frontend and database; monitoring, backup verification, and compiled final documentation.

4. Functional Requirements – Milestone 4

ID	Owner	Requirement
FR-1	Intern 1 (Backend/DB)	System shall be fully tested, hardened against bad input, and successfully deployed live so that every backend API from Milestones 1–3 works correctly on a public URL, using the unchanged dataset.
FR-2	Intern 2 (Security/Gateway)	System shall enforce correct CORS rules, secret management, and role-based access on the live deployment, verified through both automated and manual security testing.
FR-3	Intern 3 (Frontend)	System shall provide a fully deployed, responsive, error-resilient user interface connected to the live backend and AI services, with every screen tested on the public URL.
FR-4	Intern 4 (AI/ML)	System shall provide all five AI features (Forecasting, Segmentation, Churn, Recommendation, Anomaly Detection) live, tested for correctness and graceful handling of edge cases, with a documented performance summary.
FR-5	Intern 5 (Data/DevOps)	System shall be deployed end-to-end on free hosting (backend, AI services, frontend, database), monitored with a free uptime checker, backed up successfully, and accompanied by compiled final project documentation.

5. Non-Functional Requirements

- **Zero Cost:** the fully deployed platform must run entirely on free-tier tools and hosting — no paid subscription of any kind.
- **Usability:** every screen must remain just as easy to use on the live deployment as it was locally, with plain language and clear buttons.
- **Performance:** the live platform should feel responsive — pages should load and AI predictions should return within a few seconds even on free-tier hosting.

- Security: all secrets continue to live only in environment variables on the hosting platform; every endpoint keeps its correct role-based protection live, not just locally.
- Reliability: automated test suites from Milestone 3 must still pass, a live backup/restore must succeed, and a free uptime monitor must confirm the platform stays reachable.
- Data Consistency: the exact same dataset seeded in Milestone 1 (grown naturally by Invoice/Notification activity) must remain in place — no new or replacement sample dataset is introduced in this milestone.
- Scope Discipline: no new features beyond what Milestones 1–3 and the original project plan already defined; Milestone 4 effort goes entirely into testing, deployment, stability, and documentation.

6. Cross-Domain Dependency Matrix

This matrix makes sure that even though all five interns work at the same time, nobody gets stuck waiting without warning — everyone knows exactly what they need from whom, and by which day, during this short but critical 7-day milestone.

Dependency	Needed By	Why It Matters
Milestone 3 deployment drafts (Intern 5) → Actual live Backend deployment (Intern 5, with Intern 1)	By Day 3	The team executes, this milestone, the configuration that was only prepared in Milestone 3.
Live Backend URL (Intern 1 + Intern 5) → Live Frontend deployment (Intern 3, with Intern 5)	By Day 4	The Frontend cannot be meaningfully deployed until it has a real backend link to point at.
Live AI services (Intern 4 + Intern 5) → Frontend AI screens tested live (Intern 3)	By Day 5	Forecast, Segmentation, Recommendation and Anomaly screens need real, live AI responses to display.
Live Backend + Frontend (Intern 1, Intern 3, Intern 5) → Live security verification (Intern 2)	By Day 4-5	Security testing on 'localhost' is not the same as security testing on the real, public deployment.
All five live domains → Full team live integration test (Intern 5 coordinates)	By Day 5	Everyone needs to see and test the complete, real, deployed system together — not just their own piece.

7. Day-Wise Work Plan (Day 1 – Day 7)

Every day starts with a 15-minute stand-up: each intern says what they finished yesterday, what they'll do today, and any blockers. Tick each checkpoint box only when that item is actually working, not just attempted. Reminder: the same dataset from Milestone 1 is reused throughout — no new sample data, and no new features beyond what was already planned.

Intern 1 — Backend & Database Developer

Owens: Data Ingestion Layer + Storage Layer — Final Testing & Live Deployment

Day	What To Do (Explained Simply)	Checkpoints — Tick When Done
1	Recap every backend API built across Milestones 1–3 (Sales, Inventory, Invoice, Notifications, Revenue, Audit). Do a full personal test pass locally and write down any known bugs before deployment begins. Confirm the same dataset from Milestone 1 is still being used, unchanged.	☐ Full API inventory reviewed ☐ List of known bugs written down ☐ Confirmed no new or changed dataset ☐ Plan shared in the stand-up
2	Fix the critical bugs found yesterday. Add production-ready error handling everywhere — clear error messages and no server crashes, even on bad or unexpected input.	☐ Critical bugs fixed ☐ Error handling improved on all endpoints ☐ Tested with a few deliberately bad requests ☐ Backend still passes the automated tests from Milestone 3
3	Prepare the backend for deployment: move any remaining hard-coded values into the .env file, then work with Intern 5 to deploy the Backend (and its database connection) to the free cloud host (Render/Railway), using the configuration already drafted in Milestone 3.	☐ All configuration values moved into .env ☐ Backend successfully deployed to free hosting (with Intern 5) ☐ Backend live URL confirmed reachable ☐ One basic API call tested successfully on the live URL
4	Test every backend API against the LIVE deployed link (not the local copy anymore) — Sales, Inventory, Invoice, Notifications, Revenue — and fix any deployment-specific issues (wrong environment variable, CORS, database connection, etc.).	☐ All backend APIs tested on the live URL ☐ Any deployment-specific bugs fixed ☐ Live database confirmed to match the existing, unchanged dataset
5	Work with Intern 3 to confirm the deployed Frontend can talk to the live Backend correctly, and with Intern 2 to confirm RBAC and security still work correctly on the live link.	☐ Frontend-to-backend live connection confirmed working ☐ RBAC re-verified on the live link ☐ Any integration bugs fixed
6	Run a simple free load test (Apache Bench) against the live backend to see how it holds up under a small burst of requests, and write the Backend section of the final project documentation.	☐ Load test run and results noted ☐ Any performance issues addressed, or documented as a known limitation ☐ Backend documentation section written
7	Final Demo Day. Support the full live, end-to-end walkthrough with the team, fix any last-minute critical bug, and hand over clean, commented, documented backend code.	☐ Backend performs correctly during the full live demo rehearsal ☐ Any last critical bug fixed ☐ Final backend code cleaned and documented ☐ Contributed to the final presentation

Intern 2 — Security & API Gateway Developer

Owns: Application Gateway & Security Layer — Live Security Verification

Day	What To Do (Explained Simply)	Checkpoints — Tick When Done
1	Recap the full RBAC and security setup built across Milestones 1–3. List anything that might behave differently once the app is live on the internet instead of running only locally — for example CORS rules, public URLs, or secret exposure.	☐ Security setup reviewed end-to-end ☐ Local vs live' risk list written ☐ Plan shared with the team
2	Fix any security gaps found yesterday. Make sure every secret (database password, API keys) is ready to be set as an environment variable on the free hosting platform — never written directly inside the code.	☐ Security gaps fixed ☐ All secrets confirmed ready for environment-variable-based deployment ☐ .env file confirmed still in .gitignore
3	Work with Intern 5 to correctly configure CORS and allowed origins for the soon-to-be-live Frontend URL, and confirm authentication/RBAC keeps working correctly through the freshly deployed live backend link.	☐ CORS configured for the live Frontend domain ☐ Login and RBAC tested against the live backend ☐ Any access issues fixed
4	Run the automated security test suite (built in Milestone 3) against the LIVE deployed link instead of the local copy, to confirm the protections still hold in the real environment.	☐ Automated security tests run against the live link ☐ All tests passing (bad access still correctly blocked) ☐ Any live-only security issues fixed
5	Do a final manual security walk-through of the live app: try each of the four roles, try an invalid/expired token, and try direct URL access to protected pages — confirming every rule from the original Access Matrix still holds true.	☐ Every role tested on the live app ☐ Original Access Matrix rules re-confirmed on the live app ☐ Any issues found are fixed
6	Finalize the Security & Access Guide (started in Milestone 2/3) with any Milestone 4 deployment-related notes, and support Intern 1 and Intern 3 with any last integration fixes.	☐ Security & Access Guide finalized ☐ Supported teammates with integration fixes ☐ Guide shared with the whole team
7	Final Demo Day. Confirm the live app is secure and ready for the demo audience, support the full walkthrough, and contribute the security section to the final documentation and presentation.	☐ Live app confirmed secure for the demo ☐ Any last-minute fix applied ☐ Security section of the final documentation completed ☐ Contributed to the final presentation

Intern 3 — Frontend & Dashboard Developer

Owens: Presentation Layer — Deployment & Final Polish

Day	What To Do (Explained Simply)	Checkpoints — Tick When Done
1	Recap every screen built across Milestones 1–3. Do a full personal test pass and write down any UI bugs, confusing error messages, or rough edges that should be fixed before the final demo.	☐ Full screen-by-screen review done ☐ List of UI issues written down ☐ Plan shared with the team
2	Fix the UI issues found yesterday. Improve error handling on screen (clear, friendly messages instead of blank screens or technical errors), and improve responsiveness on smaller screens across the whole app.	☐ UI bugs fixed ☐ Friendly error messages added across key screens ☐ Mobile responsiveness re-checked across the whole app
3	Prepare the Frontend for deployment: move the backend URL into a configuration variable instead of hard-coding it, and run a full production build locally to confirm it completes without errors.	☐ Backend URL moved into a configuration variable ☐ Production build completed successfully with no errors ☐ Build tested locally, ready to point at the live backend
4	Work with Intern 5 to deploy the Frontend to a free hosting service (Netlify/Vercel/GitHub Pages), pointing it at the live Backend URL that Intern 1 deployed yesterday.	☐ Frontend successfully deployed to free hosting ☐ Live Frontend URL confirmed reachable ☐ Frontend correctly loads data from the live Backend
5	Test every single screen on the LIVE deployed site — Login, Dashboards, Invoice, Notifications, and all the AI-insight screens — and fix any bugs that only appear in the live environment.	☐ Every major screen tested on the live site ☐ Any live-only bugs fixed ☐ Re-tested on both desktop and mobile view
6	Do a final UI polish pass — consistent styling, loading indicators, and smooth navigation across the whole live app — and take final screenshots for the project documentation.	☐ Final polish pass completed on the live site ☐ Loading indicators present wherever data is being fetched ☐ Screenshots taken for the final documentation
7	Final Demo Day. Rehearse and support the full live click-through demo, fix any last-minute UI bug, and contribute the Frontend section to the final documentation.	☐ Full live demo walkthrough rehearsed successfully ☐ Any last-minute UI bug fixed ☐ Frontend documentation section completed ☐ Contributed to the final presentation

Intern 4 — AI/ML Engineer

Owens: AI Analytics Engine — Final Testing & Live Deployment of All 5 Models

Day	What To Do (Explained Simply)	Checkpoints — Tick When Done
1	Recap all five AI features built across Milestones 2–3 (Sales Forecasting, Customer Segmentation, Churn Prediction, Product Recommendation, and Anomaly Detection). Re-test each one locally on the existing dataset and record its current metrics (MAE/RMSE for forecasting, accuracy/precision/recall for churn, silhouette score for segmentation, and so on, as defined in the original project plan).	☐ All 5 AI features re-tested locally ☐ Current metrics recorded for each (using the existing dataset only) ☐ Any obvious issues listed
2	Fix any issues found yesterday. Make sure every AI feature handles a tricky or edge-case input gracefully — for example a brand-new customer with no purchase history — instead of crashing.	☐ Issues fixed ☐ Edge cases tested for each AI feature ☐ No crashes found on edge-case inputs
3	Package all five AI services cleanly, using the same Flask/FastAPI structure already built in Milestone 2/3, so they are ready to deploy — then work with Intern 5 to deploy them to the free cloud host.	☐ AI services packaged and deployment-ready ☐ AI services successfully deployed (with Intern 5) ☐ Live AI service URLs confirmed reachable
4	Test all five AI features against their LIVE deployed links, and fix any deployment-specific issues — for example a missing model file, a slow cold start, or a memory limit on the free hosting tier.	☐ All 5 AI features tested on their live links ☐ Any deployment-specific bugs fixed ☐ Response times checked and noted
5	Work with Intern 3 to confirm the Frontend's AI-related screens (Forecast vs Actual, Customer Insights, Recommendations, Anomaly Alerts) correctly display data coming from the LIVE AI services.	☐ All AI-related Frontend screens tested against the live AI services ☐ Any data-format mismatches fixed ☐ Confirmed with Intern 3 that everything displays correctly
6	Write a short, simple 'Model Performance Summary' listing the final metrics for each of the five AI features, matching the evaluation categories defined in the original project plan, and support any last testing.	☐ Model Performance Summary written for all 5 features ☐ Metrics match the categories defined in the original project plan ☐ Document shared with the team
7	Final Demo Day. Support the full live demo showing all five AI features working correctly on real data, fix any last-minute AI bug, and contribute the AI/ML section to the final documentation.	☐ All 5 AI features demonstrated successfully live ☐ Any last-minute bug fixed ☐ AI/ML documentation section completed ☐ Contributed to the final presentation

Intern 5 — Database & DevOps Engineer

Owns: Storage Layer + Infrastructure Layer — Executing the Zero-Cost Deployment

Day	What To Do (Explained Simply)	Checkpoints — Tick When Done
1	Recap the deployment configuration that was drafted (but intentionally not executed) back in Milestone 3. Confirm the free-tier accounts (Render/Railway, Netlify/Vercel, the existing free database) are ready, and agree the exact deployment order for this milestone with the team.	☐ Milestone 3 deployment drafts reviewed ☐ Free-tier accounts confirmed ready ☐ Deployment order/plan shared with the team
2	Finalize the shared free cloud database connection that was already set up in Milestone 2/3. Confirm it still holds the exact same dataset as before, and is ready and reachable for the live deployment.	☐ Free cloud database confirmed reachable ☐ Dataset confirmed unchanged from previous milestones ☐ Connection details ready to share securely with Intern 1
3	Execute the deployment of the Backend and the five AI services to the free cloud host (Render/Railway), working closely with Intern 1 and Intern 4.	☐ Backend deployed and reachable via a live URL ☐ AI services deployed and reachable via live URLs ☐ All services connected correctly to the shared cloud database
4	Execute the deployment of the Frontend to a free hosting service (Netlify/Vercel/GitHub Pages), working closely with Intern 3, and confirm it is correctly pointed at the live Backend URL.	☐ Frontend deployed and reachable via a live URL ☐ Frontend correctly connected to the live Backend ☐ Full live link shared with the whole team
5	Coordinate a full live integration test with the whole team — every domain (Backend, Security, Frontend, AI) tested together on the actual live links, not locally — and log any bugs for the right teammate.	☐ Full live integration test completed with the team ☐ Bugs logged and assigned to the correct teammate ☐ Test results shared with the team
6	Set up simple free monitoring on the live deployment (UptimeRobot) and confirm the backup process from Milestone 2/3 still works correctly against the live cloud database.	☐ Free uptime monitor set up on the live backend and frontend ☐ Backup taken and restore tested against the live cloud database ☐ Any issues found are fixed
7	Final Demo Day. Do one final full check of the live deployed platform, compile the final project documentation from all five interns into a single document, and support the final live demonstration.	☐ Final full live-platform check completed ☐ Final project documentation compiled from all five sections ☐ Live demo supported successfully ☐ Any last-minute deployment issue fixed

8. Daily Synchronization Protocol

8.1 Daily Stand-up (15 minutes, every morning)

- Each intern answers: What I finished yesterday / What I will do today / Any blockers.
- Mentor/Team Lead checks the Dependency Matrix (Section 6) to confirm no one is stuck — with only 7 days, blockers must be caught immediately.

8.2 Mid-Point Mini-Integration Check (Day 4 evening)

- Intern 5 confirms Backend, AI services and Frontend are all live and reachable by the whole team.
- Intern 2 confirms security and RBAC are behaving correctly on the live deployment, not just locally.

8.3 Shared Artifacts (must be updated daily)

- Shared checklist/task board showing each intern's daily checkpoints (a free tool like Trello or a shared Google Sheet works fine).
- A shared document with all live URLs (Backend, Frontend, AI services) so every intern always has the current links.
- Shared Git repository, with each intern's own feature branch merged into a common integration branch by Day 6.

9. Day 7 Integration Plan & Acceptance Criteria

Day 7 is fully reserved for the final live demonstration — no new features or risky changes should be attempted on this day.

9.1 Integration Sequence

- Step 1: Intern 5 confirms the live Backend, live AI services, live Frontend and cloud database are all reachable and connected together.
- Step 2: Intern 2 re-confirms RBAC and security hold correctly on the live public link, for all four roles.
- Step 3: Intern 3 walks through every Frontend screen on the live site — Login, Dashboards, Invoice, Notifications, and all AI-insight screens.
- Step 4: Intern 4 demonstrates all five AI features live, using the same original dataset, and shows the final performance metrics.
- Step 5: Intern 1 confirms Sales, Inventory, Invoice, Notifications and Revenue Summary all behave correctly on the live link.
- Step 6: The whole team walks through the complete Business Owner, Store Manager, Sales Executive and Administrator journeys together, live, and fixes any last defects.

9.2 Milestone 4 Acceptance Criteria

- The entire platform is reachable through a single, free, public web link that anyone on the team (or a reviewer) can open (Intern 5).
- All four roles can log in and see the correct role-based dashboard and permissions live, matching the original Access Matrix (Intern 1 + Intern 2).
- Sales upload, inventory tracking, invoice management, and notifications all function correctly on the live link (Intern 1).
- All five AI features (Forecasting, Segmentation, Churn, Recommendation, Anomaly Detection) return correct, real results on the live link, using the same original dataset (Intern 4).

- The Frontend is responsive, handles errors gracefully, and every screen works correctly on the live deployment (Intern 3).
- Security and RBAC are verified working correctly on the live, public deployment — not only in local testing (Intern 2).
- A free uptime monitor is actively watching the live platform, and a backup of the live database has been successfully taken and restored at least once (Intern 5).
- No new or replacement dataset has been introduced anywhere in the project — the original Milestone 1 dataset remains the single source of truth throughout (All interns).
- Final project documentation (covering architecture, modules, AI performance, security, and deployment) is compiled and complete (All interns, compiled by Intern 5).
- The whole team has successfully rehearsed and delivered a full, live, end-to-end demonstration of MarketMind AI (All interns).

10. Risks & Mitigation

Risk	Mitigation	Owner
Free hosting tier is slow to 'wake up' (cold start) during the live demo	Test response times early (Day 3-4); 'warm up' the live links a few minutes before the actual demo.	Intern 5
Live deployment breaks something that worked perfectly locally	Deploy early (Day 3-4), not on the last day, so there is time left to find and fix issues.	Intern 5 / All
CORS or environment-variable misconfiguration blocks live login	Test immediately after each deployment step, not only at the very end of the milestone.	Intern 1 / Intern 2
Free hosting service has a temporary outage right before the demo	Keep the Milestone 3 local docker-compose version fully working as a backup demo option.	Intern 5
Only 7 days available, risk of trying to add 'just one more feature'	Strictly limit scope to testing, deployment, stability and documentation — no new functionality this milestone.	All / Mentor
Someone accidentally changes or replaces the shared dataset during final testing	Freeze the dataset on Day 1; any data question goes through Intern 5 first.	All / Intern 5

11. Sign-Off

This SRS for Milestone 4 — the final milestone — is to be reviewed and acknowledged by all five interns and the mentor/educator before work begins on Day 1.

- Mentor / Educator: _____ Date: _____
- Intern 1: _____ Date: _____
- Intern 2: _____ Date: _____
- Intern 3: _____ Date: _____
- Intern 4: _____ Date: _____
- Intern 5: _____ Date: _____